

Branch Specialization Checkpoint

Unlabeled Router Regularization

Branch Specialization Project

2026-05-22

Checkpoint reason: generic gate regularizers did not reliably induce causal modularity.

Previous checkpoint:

- weak role-label routing at weight 0.05 produced 5/5 causal modularity;
- gate compliance appeared before causal modularity.

This checkpoint asks:

Can unlabeled routing pressure produce similar branch-level functional modularity without local/induction role labels?

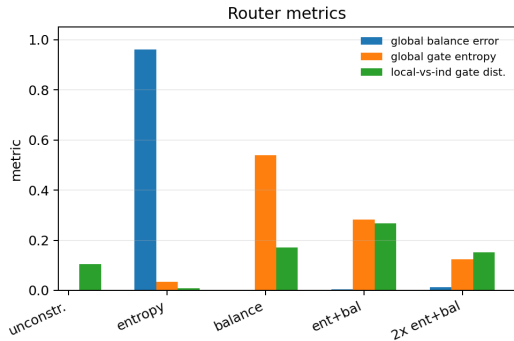
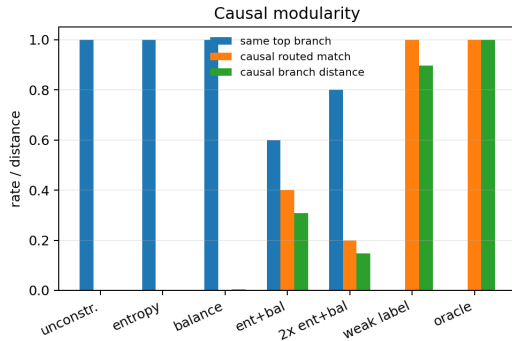
Same local-vs-induction competition task:

- config: `learned_token_router`;
- no scored-position role-label routing loss;
- five seeds per condition, 1200 training steps;
- branch modularity measured by branch ablation.

Unlabeled regularizers:

- entropy minimization: make per-position gates sharper;
- load balancing: keep global branch usage near 50/50;
- entropy plus load balancing.

Main Evidence



Entropy and balance regularizers change router statistics, but none approaches the weak-label or oracle causal modularity result.

Causal Modularity

Condition	Local acc.	Ind. acc.	Same top	Routed match	Branch dist.
unconstrained	1.0000	0.9989	1.00	0.00	0.0006
entropy	1.0000	0.9994	1.00	0.00	0.0002
balance	1.0000	0.9993	1.00	0.00	0.0033
ent+bal	1.0000	0.9993	0.60	0.40	0.3082
2x ent+bal	1.0000	0.9984	0.80	0.20	0.1484
weak label	0.9998	0.9983	0.00	1.00	0.8957
oracle	1.0000	0.9988	0.00	1.00	1.0000

All unlabeled conditions solve the task, but remain far below weak labels on causal routed match.

Router Metrics

Condition	Balance error	Global entropy	Role gate dist.	Gate match
entropy	0.9597	0.0351	0.0095	0.00
balance	0.0017	0.5385	0.1718	0.00
ent+bal	0.0038	0.2830	0.2667	0.40
2x ent+bal	0.0118	0.1247	0.1512	0.20

Load balancing can be almost perfect while local and induction remain functionally co-located.

Entropy-only:

- gates become sharp;
- each seed collapses both roles onto the same branch.

Balance-only:

- global branch usage is almost exactly 50/50;
- local and induction still depend on the same top branch in 5/5 seeds.

Entropy plus balance:

- sometimes finds a role split;
- remains mixed and far below weak labels.

Supported in this toy setup:

- Sharp routing is not functional modularity.
- Balanced routing is not functional modularity.
- Generic unlabeled gate regularizers are insufficient here.

Current formulation:

Structural branches and learned gates provide a substrate for modularity, but functional modularity needs role-informative routing pressure or equivalent task pressure.

Stop sweeping generic gate regularizer weights on the current easy task.

Next decisive experiment:

Increase role conflict or impose branch bottlenecks, then test whether unlabeled routing pressure begins to align with functional roles.

The next result should distinguish whether the failure is due to the regularizer class or due to the task allowing cheap cohabitation.

Sources and Provenance

- Report: `doc/phase3_toy_unlabeled_router_regularization.md`
- Model script: `scripts/toy_branch_isolation_intervention.py`
- Analysis script: `scripts/analyze_unlabeled_router_regularization.py`
- Analysis output: `results/phase3_toy_unlabeled_router_regularization_analysis/`
- Deck data snapshot: `presentations/2026-05-22-1551-unlabeled-router/data/`